

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

DEVELOPMENT AND IMPLEMENTATION OF A ROBOTIC APPLICATION FOR HUMAN INTENTION RECOGNITION USING MOTION AND HAND GESTURE

*A Final Project Report Submitted in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Computer Engineering*

PRESENTATION BY:

ELEANA OSEI OWUSU — 4121230024

JOEL NII ADJETEY AHULU — 4121230020

SUPERVISOR: MR. MICHEAL XENYA

ACADEMIC YEAR: 2025/2026

PRESENTATION OUTLINE

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- 03. System Architecture & 6-Phase Methodology**
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01. INTRODUCTION & RESEARCH BACKGROUND

Robots Working Directly with Humans

Robots are leaving safety cages to work alongside people in hospitals, warehouses, and homes. For humans and robots to coexist safely, the robot must understand where people are and respect their personal space.

Natural, Hands-Free Interaction

Instead of requiring handheld controllers, touchscreens, or keyboards, humans can guide the robot using natural hand gestures and body movement—just like signaling to a human coworker.

Fast Onboard Intelligence (No Cloud Lag)

Sending live video over the internet causes dangerous delays (>200 ms) and fails when Wi-Fi drops. In this project, all AI recognition and navigation decisions happen entirely inside the robot in real time.



Figure 1.1: Collaborative Human-Robot Interaction (HRI) in an industrial shared workspace, illustrating contactless gesture coordination.

02. PROBLEM STATEMENT

Current mobile robots separate vision recognition from active navigation, relying on remote cloud servers that suffer from high network lag (>200 ms) and connection dropouts.

In shared human-robot workspaces, this delay prevents robots from reacting in real time to human gestures, creating severe collision hazards under ISO 15066 safety standards.

PERCEPTION-ACTION GAP

Vision AI isolated from wheel motors & obstacle avoidance
(Tsitos et al., 2022)

CLOUD LATENCY HAZARD

Network lag > 200 ms & Wi-Fi drops
violate ISO 15066 safe stopping
margins

EDGE COMPUTING LIMIT

Heavy vision AI overheats mobile chips; demands a sub-135 ms
onboard pipeline



03. PROJECT OBJECTIVES

GENERAL OBJECTIVE

To design, implement, and evaluate an edge-deployed, real-time human intention recognition and autonomous navigation system powered by ROS 2 on an embedded mobile robot.

01. Invariant Vision Pipeline

Compute 19 scale-invariant geometric features from 21 hand landmarks for real-time onboard MLP classification.

02. Balanced 6,000-Sample Dataset

Capture and balance a custom 6-class dataset directly through the onboard camera to eliminate perspective domain shift.

03. LSTM Trajectory Prediction

Forecast short-horizon operator motion trajectories from continuous body pose landmarks using recurrent neural networks.

04. Supervisory Brain Node

Execute a deterministic finite state machine with 5-frame rolling consensus, active operator lock, and visual servoing.

05. 2D LiDAR SLAM & Nav2

Deploy real-time Cartographer mapping and the Nav2 navigation stack for socially aware, collision-free path execution.

06. Empirical Safety Benchmarking

Quantify end-to-end latency (132 ms), physical classification accuracy (96.7%), and compliance with ISO 15066 safety margins.



04. LITERATURE REVIEW & RESEARCH GAPS

BENCHMARK LITERATURE EVALUATION

Tsitos et al. (2022)

Kinematic Intention on Robotic Arm

[+] Proved sub-150 ms latency threshold on industrial UR3 manipulator.

[-] Confined to stationary tabletop; zero mobile navigation or explicit gestures.

Mahmud et al. (2023)

3D Skeletal Tracking & Gesture AI

[+] High multi-gesture classification accuracy using deep vision models.

[-] Relies on heavy GPU / cloud servers; vulnerable to network lag & dropouts.

Li et al. (2024)

Predictive Costmaps & ROS 2 Nav2

[+] Adapted ROS 2 costmaps dynamically based on obstacle movement.

[-] Simulation-only (Isaac Sim); unverified on physical embedded edge hardware.

CRITICAL RESEARCH GAPS RESOLVED IN THIS WORK

01. Edge-Native Processing

Eliminates remote cloud latency (>200 ms) and connection dropouts by executing the entire vision pipeline onboard an embedded Raspberry Pi 5.

02. Perception-Action Coupling

Bridges isolated vision AI to real-time wheel actuation via ROS 2 DDS middleware, active visual servoing, and a 5-frame rolling consensus FSM.

03. Dual-Modality HRI & Safety

Unifies discrete hand gestures with continuous LSTM trajectory prediction and 2D LiDAR SLAM, strictly complying with ISO 15066 safety stopping margins.

